Retrocomputing as an Integral Part of Digital Humanities Practice?

Roeder, Torsten
torsten.roeder@uni-wuerzburg.de
Universität Würzburg, Germany

Digitality permeates and shapes contemporary everyday life. However, it is often forgotten that our current understanding of digitality is the result of a long developmental process and, therefore, has a far-reaching historical dimension. While the term "digitization" has only recently gained prominence as a political buzzword, the underlying techno-social transformation has been underway for more than half a century. By the 1970s, with the proliferation of game consoles, and the 1980s, with the advent of home computers that easily integrated into existing household infrastructures (e.g., by using televisions as screens and cassette recorders for loading and storing programs), digitization had already reached broader segments of the population.

Retrocomputing shifts the focus to older computer systems that have been overshadowed by the logic of acquisition and disposal inherent in the computer industry and are now traded as vintage items. Regardless of whether it is an iPhone 14 from 2023 or an Apple II from 1977, computers remain computers and, in principle, can perform the same tasks as envisioned by Turing and von Neumann. However, they are subject to a dynamic historical development that cannot be fully understood solely through the empirically observable increase in processing power (Moore's Law). It must also take into account the diversity of interfaces as well as purpose-driven and target-specific configurations. These devices also include the software executed on them, encompassing the programming languages and techniques used. From this perspective, it becomes clear that earlier concepts and visions of digitality were fundamentally different from today's. At the same time, cultural markers and habitual patterns emerged that persist in some cases to this day. Consider, for instance, the character Donkey Kong or the row-column syntax in spreadsheets, both created or established around 1980. Today's digital culture cannot be fully understood or explained without incorporating its earlier forms and structures.

These historical forms of digitality are the focus of the Retro Computing Lab at the Center for Philology and Digitality (ZPD) at the University of Würzburg (fig. 3). There, the study of old computer systems and their associated games and applications is not conducted through emulators but rather on the original devices. By late 2024, about 15 legacy systems (both home computers and game consoles) will be operational there. The setups, including CRT monitors, floppy drives, and dot-matrix printers, not only offer an authentic approach to hardware and software but also enable an understanding of the historical context in which games and applications were developed, forming the multimedia backdrop of current retro games. Furthermore, they allow access to digital legacies such as unpublished games and their source codes (see Games That Weren't or the NZTronix Collection). This provides students from various disciplines, ranging from computer science to German studies to museology, with expanded opportunities to engage with historical forms of digitality and digital heritage. The location at the Center for Philology and Digitality also integrates this into the bridging (study) disciplines of Digital Humanities and Computational Humanities.

Similar collections and projects can be found at various locations. Notable examples include MITH's Vintage Collection (University of Maryland) and the Media Archaeology Lab (University of Colorado). In Europe, there are also numerous collections, though often outside the scope of Digital Humanities, typically implemented as museums and much less frequently as an integral part of an educational institution or, let alone, a DH institute. This raises the question of whether and to what extent retrocomputing is part of the Digital Humanities.

To address this, I propose four aspects for discussion:

Self-Reflection of DH in the Context of Digitality History

A historical perspective shows that the Digital Humanities primarily operate adaptively, adopting new developments in IT and tailoring them for specific applications in the humanities and social sciences. This forward-looking approach results in high dynamism but also leaves behind a trail of obsolete technologies. The Digital Humanities, whose historiography currently begins with the use of punch cards, are thus themselves subject to an ever-changing digitality. It seems only logical, as part of a professional self-reflection, to engage with the evolution of digitality. Even if computers, as mentioned earlier, remain fundamentally computers, the specific manifesta-

tions of interfaces, data structures, and processing methods significantly shape the digital artifacts left to posterity. This also continually confronts the Digital Humanities with models that must operationalize the complex realities of humanities inquiries.

"Computativity" as Part of DH Didactics

The Digital Humanities could also be embedded in a general cultural history of digital media. However, this perspective might obscure a core constant of DH: the challenge of developing solutions for modeling, analysis, and presentation methods with the computational tools and digital infrastructures available at a given time and place. Problems encountered in representing historical or literary models using text or database models in 1975 could arise similarly in 2025 when continuous information must be translated into a system inherently based on discrete units. Computers, after all, remain computers. Engaging with historical digitalities can thus serve as an abstraction process, revealing fundamental questions of the Digital Humanities independently of their specific or current implementation conditions. Therefore, retrocomputing should be considered not only a historical but also a didactic component of the Digital Humanities.

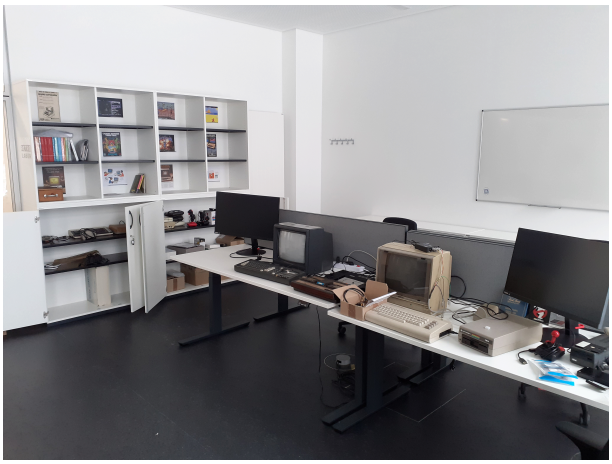

Fig. 1: Coding practices of the 1980s, type-in programs in computer journals.

Legacy Computing as a FAIR Measure

Beyond self-reflective and didactic uses, retrocomputing offers further possibilities. While "born-digital" objects are increasingly produced, much legacy data is lost due to technological progress, as current software may no longer interpret it, or the conditions of its creation are no longer comprehensible. Retrocomputing could reconstruct original setups to make data visible in their original context, potentially in an emulated environment, and migrate data to interoperable formats with proper documentation. This would help ensure that older and future datasets remain accessible within the FAIR cycle (cfr. Rinne 2011; Feichtinger 2023). Given the technological diversity, it is efficient for retrocomputing centers to specialize in specific areas, such as mainframes, home computers, Windows systems, or Apple systems.

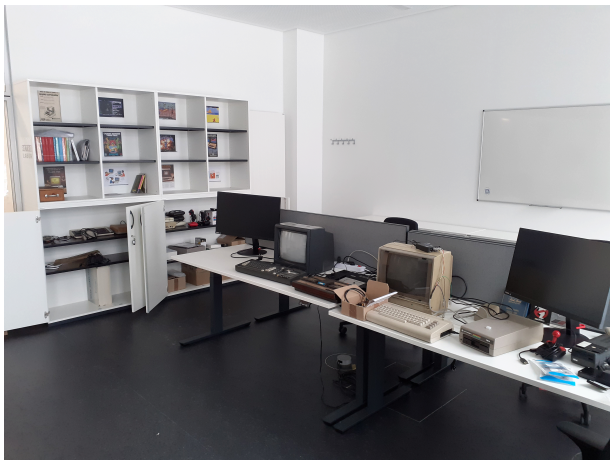

Fig. 2: A Schneider CPC 464 (1984) and data recovered from 3" floppy disks.

Digital Heritage Preservation as Part of Institutional Expertise

Preserving earlier digital systems as operational cultural objects will become increasingly important as future generations produce more digital cultural heritage. Instead of migrating data and removing it from its original context, digital objects could be maintained in their original or emulated environments (cfr. Acker 2021; Schiller/Grigar 2019). However, this requires a high degree of

specialization and methodological groundwork at the intersection of DH and computer archaeology (cfr. Schröter 2023) to establish practical and scientific standards.

Outlook

The Retro Computing Lab at the University of Würzburg, still in its infancy, aims to address these aspects in the future. How these can be implemented in practice—through research projects, courses, public outreach, and collaboration with other disciplines such as computer science, media studies, cultural studies, and more—will be revealed over time, as these fields increasingly engage with digital data and cultural heritage.

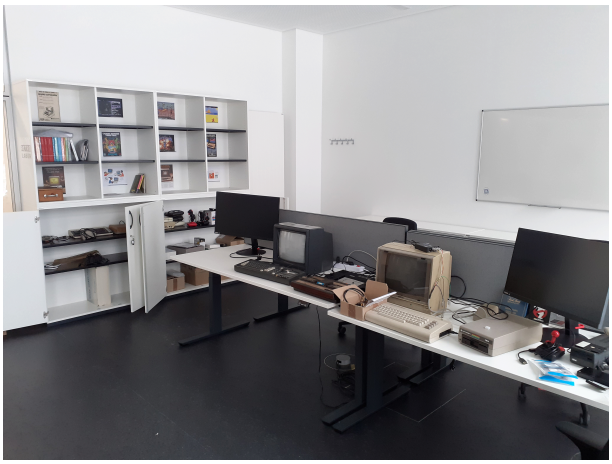

Fig. 3: The Retro Computing Lab at the University of Würzburg.

Bibliography

Amelia Acker: Emulation Practices for Software Preservation in Libraries, Archives, and Museums. In: Journal of the Association for Information Science and Technology 72, 2021, Nr. 9, S. 1148–1160, https://doi.org/10.1002/asi.24482

Moritz Feichtinger: Annotation, Simulation und Analyse eines historischen Datenbanksystems. In: Informatik 2023 – Designing Futures: Zukünfte gestalten. Bonn, Berlin 2023, S. 863–872, https://doi.org/10.18420/INF2023_96

Thorsten Ries: Das digitale ‹dossier génétique›. Digitale Materialität, Textgenese und historisch-kritische Edition. In: Textgenese in der digitalen Edition. Hrsg. von Anke Bosse und Walter Fanta. Berlin, Boston 2019 (Beihefte zu editio. 45), p. 91–116, https://doi.org/10.1515/9783110575996-007

Christoph Rinne: Digitale Archäologie. Nachlass von Jürgen Hoika †. In: Journal of Neolithic Archaeology 13, 2011, https://doi.org/10.12766/jna.2011.62

Nicholas Schiller, Dene Grigar: Born Digital Preservation of E-Lit: A Live Internet Traversal of Sarah Smith's King of Space. In: International Journal of Digital Humanities 1, 2019, Nr. 1, S. 47–57, https://doi.org/10.1007/s42803-019-00004-w

Jens Schröter: Medienarchäologie der digitalen Medien. In: Handbuch Digitale Medien und Methoden. Hrsg. von Sven Stollfuß, Laura Niebling und Felix Raczkowski. Wiesbaden 2023, https://doi.org/10.1007/978-3-658-36629-2_21-1

Sydney J. Shep: Digital Materiality. In: A New Companion to Digital Humanities. Hrsg. von Susan Schreibman, Ray Siemens und John Unsworth. Hoboken 2016, S. 322–330